



UNIVERSITÀ
DI TRENTO

Software Engineering

Design Thinking

25/02/2026

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a.a. 2025-2026

Course objective

The objective of the course is about providing the students concepts, approaches and techniques for the analysis, the design, and the development of the software

Outline

- Design thinking
 - Intro & origin
 - Definition
 - Applications
 - Solution-based thinking
 - Analysis vs Synthesis
 - Divergent & convergent thinking
 - Attributes
 - Process (Empathize, Define, Ideate, Prototype, Test)
- Design thinking and IT industry the agile era

Class material

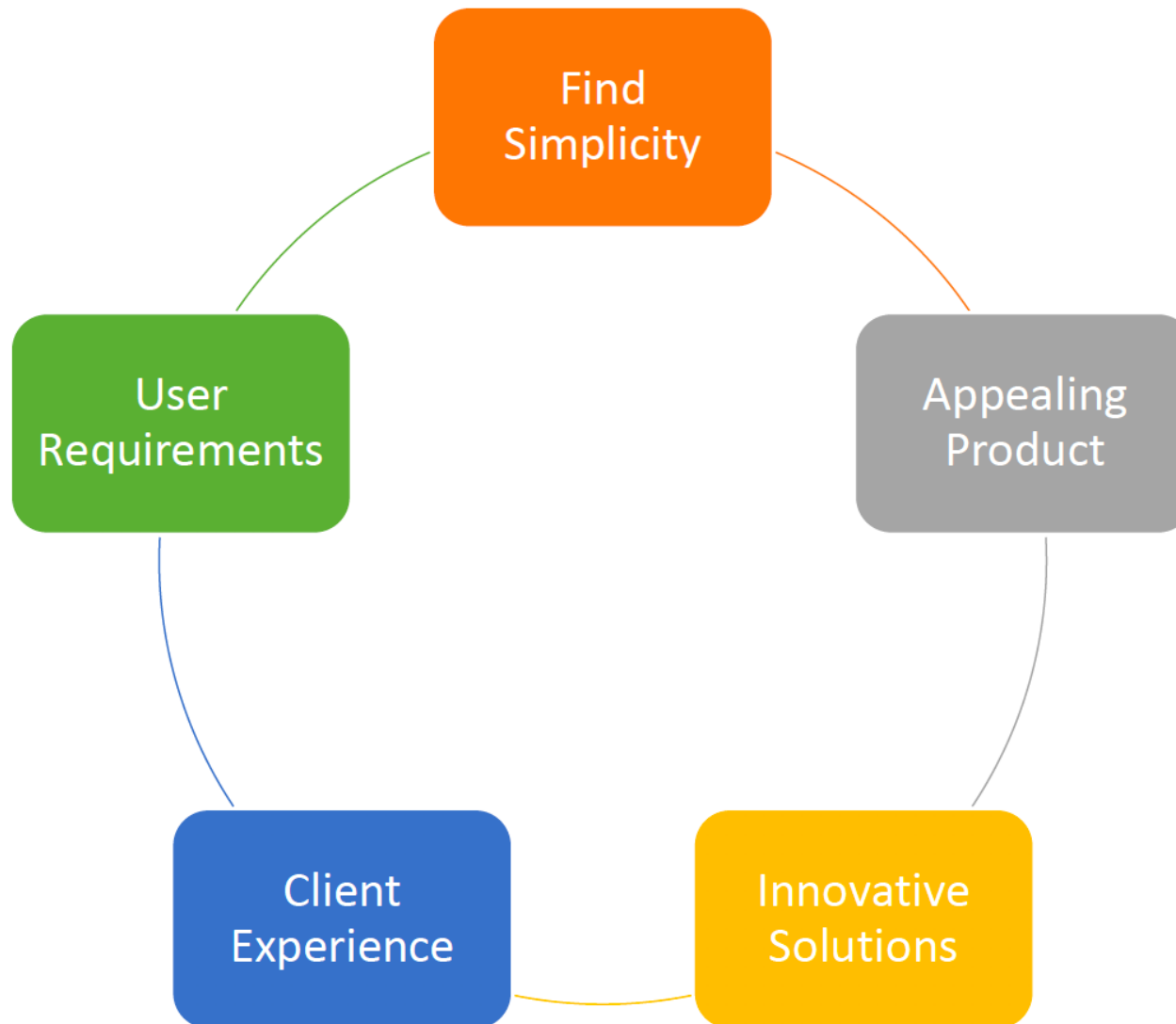
- Tutorial (chapter 1 & 2):
https://www.tutorialspoint.com/design_thinking/design_thinking_tutorial.pdf
 - Full tutorial at <https://store.tutorialspoint.com/>
- An introduction to design thinking PROCESS GUIDE
<https://web.stanford.edu/~mshanks/MichaelShanks/files/509554.pdf>
- Credits: slides cover all the relevant chapters of the tutorial (1-13,17,19) + PROCESS GUIDE

Let's now focus on
design aspects

Design thinking

- Design thinking is a **methodology** that designers use to brainstorm and solve complex problems related to **designing** and **design engineering**
- It is also beneficial for designers to find **innovative, desirable** and **never-thought-before** solutions for **customers** and **clients**
- It is used extensively in the area of **healthcare** and **wellness, agriculture, food security, education**, financial services, and environmental sustainability, etc.

Features of design thinking (I)



Features of design thinking (II)

*Design thinking is a methodology for finding **simplicity in complexity**, improving **quality of experience** with the **designed products** and **serving the needs of customers** by addressing the **target problem** faced by them*

Features of design thinking (III)

- Most of the challenges in the world do not get solved because people trying to address those problems focus too much on the **problem statement**
- *Design thinking helps to gain a balance between the **problem statement** and the **solution developed***
- **A design-oriented mindset** is not problem focused, but **solution focused**. It has to involve both **analysis** and **imagination**
- Design thinking is the way of resolving issues and dissolving problematic situations by the **help of design**

Design thinking & innovation

- Considered to be a **strategy for innovation**
- It is at the core of **effective strategy development** and **seamless organizational change**
- **Anything that involves human interaction, from products, services, processes etc., can be improved through design thinking**

Use of Design Thinking

- The **basic principle** of design thinking is that **innovation can be disciplined**
 - **Innovation** is a practice that can be **systematically approached** by a set of practical and meticulous **tools, methodologies, and frameworks**

Design thinking helps you learn the following

- How to **optimize** the ability to innovate?
- How to develop a variety of concepts, products, services, processes, etc. **for end users**?
- How to leverage the **diverse ideas** of innovation?
- How to convert useful data, individual insights and vague ideas into **feasible reality**?
- How to connect with the customers and end-users by targeting their **actual requirements**?

Design thinking helps in

Optimization
of
capabilities

Feasibility
and viability
analysis

Addressing
the needs of
customers

Applications (I)

- Design thinking is halfway between **analytical thinking** and **intuitive thinking**
- Two extremes:
 - Analytical thinking involves purely **deductive reasoning** and **inductive logical reasoning** that utilize **quantitative** methodologies to come to conclusions.
 - Intuitive thinking refers to **knowing something without any kind of reasoning**
- The intuitive thinking helps in **invention for the future**, whereas analytical thinking to create **something creative in the present**, which is **replicable**.

Applications (II)

- **Business**
 - optimization of the product creation process
 - customer → deep empathy + users needs
- **Information technology**
 - It is not only about tech or products but also processes
- **Education**
 - How to solve the problems students are facing in the classroom?
- **Healthcare**
 - Healthcare expenses are continuously growing
 - How to bring quality healthcare to people at low cost?

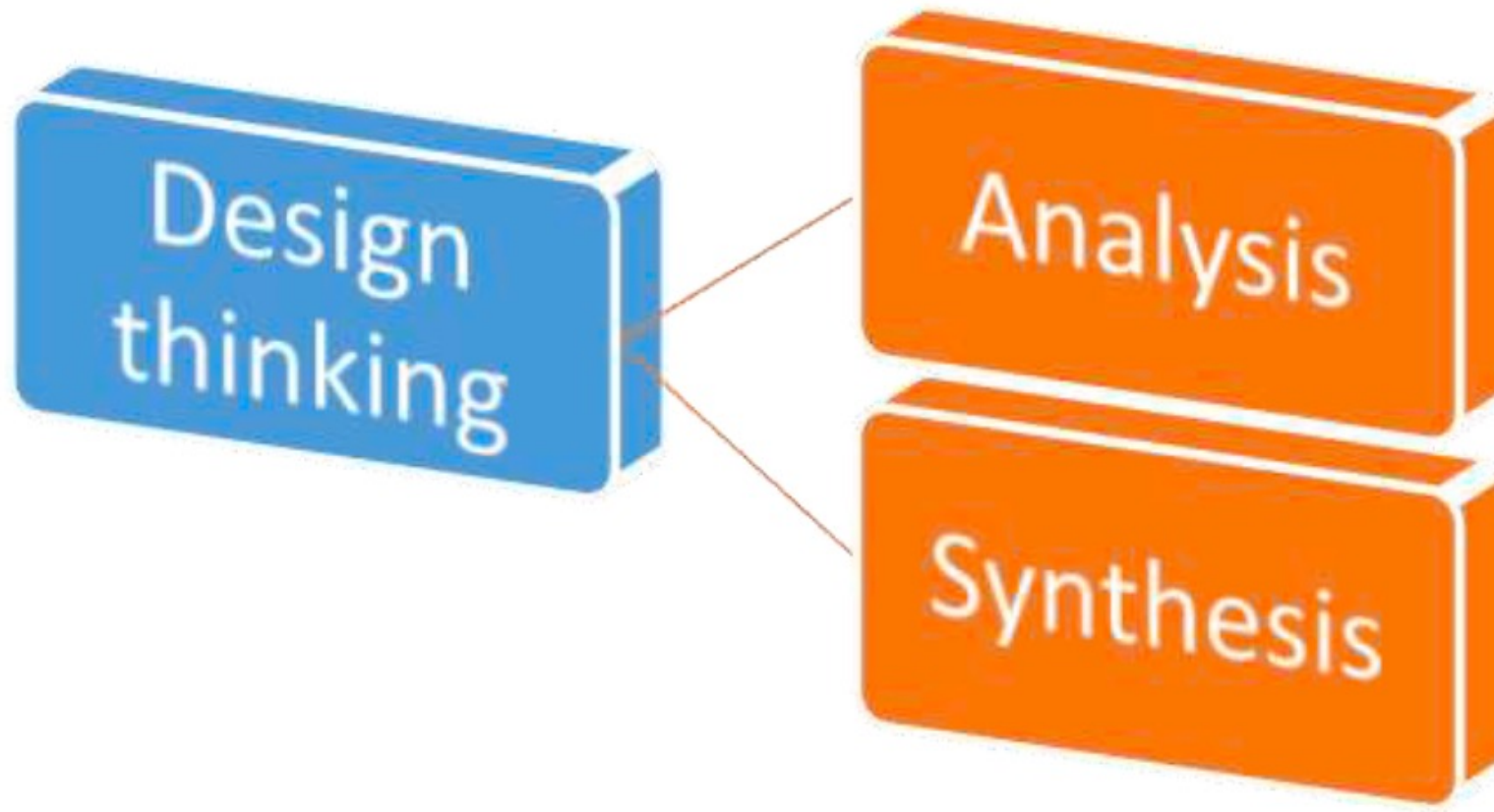
Solution-based thinking

- Design thinking provides **systematic approach to finding solutions**
 - It encourages **solution focused/based thinking**
- The design thinker is supposed to have a **clear idea of the goal** of the entire process
- This methodology helps because **alternative solutions** can be explored **simultaneously**.
 - by focusing on both the present and the future conditions as well as the parameters of the problem statement

Design thinking vs Scientific method

- The scientific method begins with **rigorously defining all the parameters of the problem**, so as to arrive at a solution.
- A design thinker is supposed to identify both the **known** and the **ambiguous facets of the problem statement** along with the current situation.
- This method of thinking helps to **unearth hidden parameters** and **open alternate paths** to reach the solution.
- Design thinking is iterative, which can also lead to redefining the problem statement

Design thinking = Analysis + Synthesis



Analysis vs Synthesis

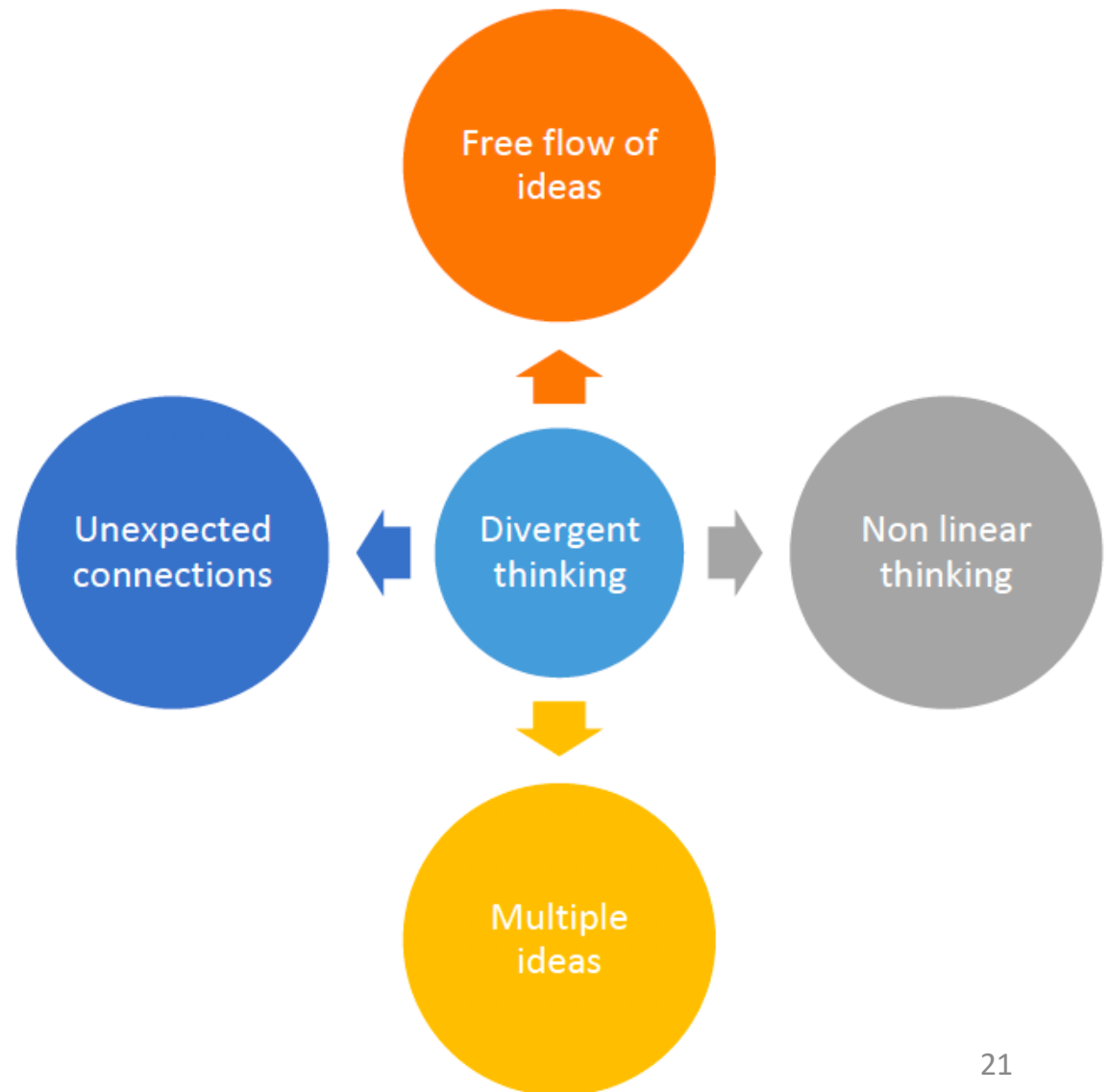
- Analysis refers to the process of **breaking down something** substantial into **multiple fragments** or components.
 - Each component are to be solved one-by-one
 - Reductionism (*“the whole is equal to the sum of its parts”*)
 - Later, a feasibility check removes non feasible/viable solutions
- Synthesis is the total contrast of analysis. In Synthesis, we **combine fragmented elements** to form an aggregated and **coherent whole**
 - Done at the end of the creative process

Divergent and Convergent Thinking

- Design thinking involves two types of thinking:
 - Divergent
 - Convergent

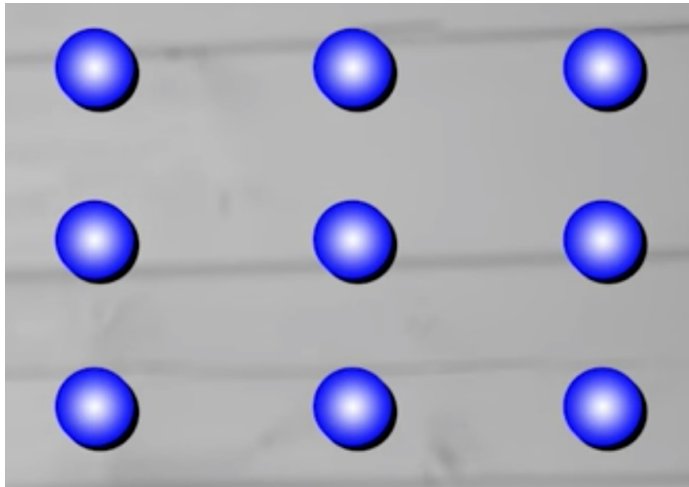
Divergent thinking

- Divergent thinking is the process of devising **more than one solution** for a problem statement
- It refers to the thought process of **generating creative solutions**
- **No judgment**



Creativity exercise: 9 points problem

Connect the 9 dots with 4 straight lines without the pen leaving the paper



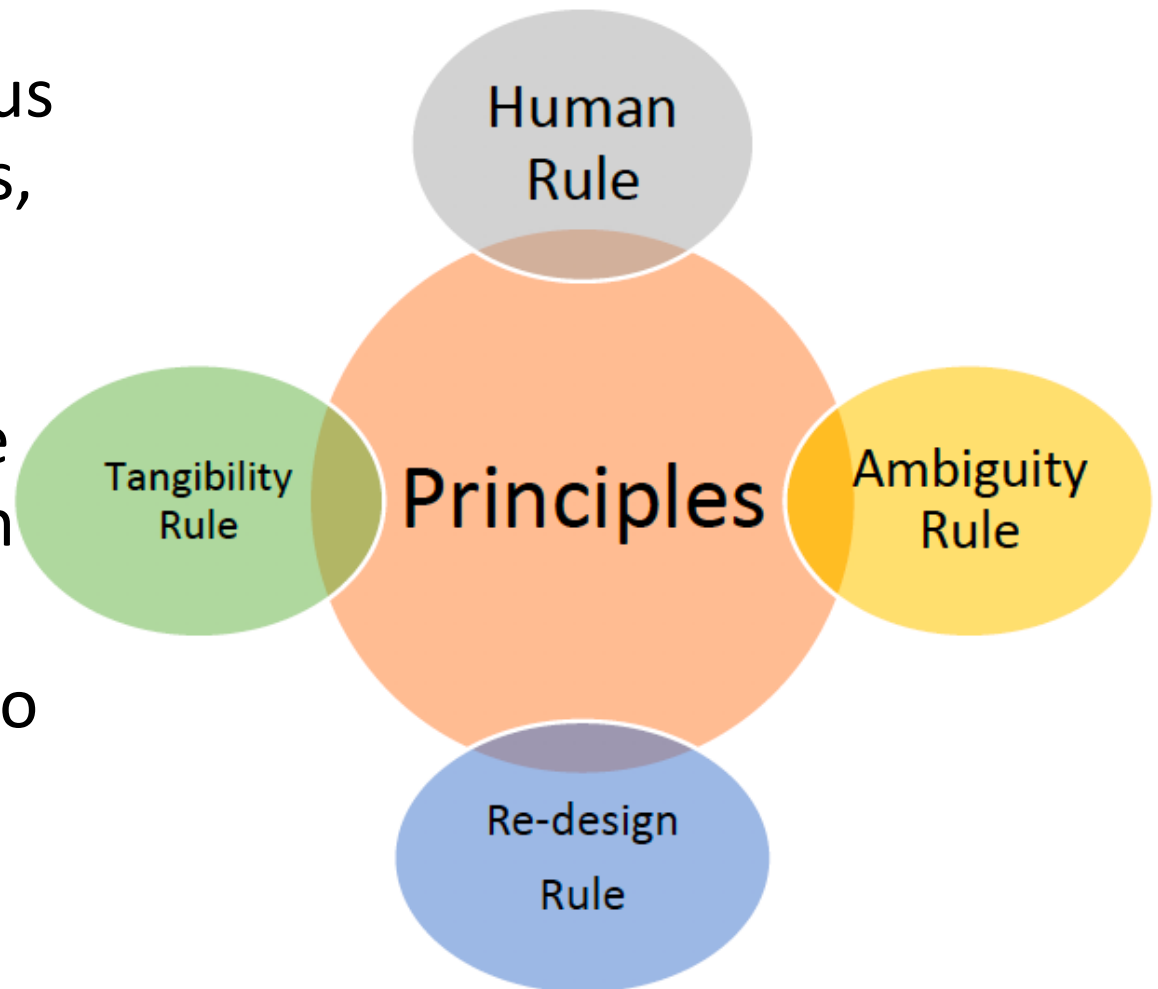
Convergent thinking

- It requires the design thinker to go through all the possible solutions thought during divergent thinking and come up with a correct solution
- This convergence on a single solution (or a mix of limited number of solutions) is the **essence of convergence thinking**
- it should help us arrive at a single **best answer without any room for ambiguity**
- **Judgment is an important** part of this process.

Attributes (I)

Principle of Design Thinking

- Design thinking is an extensive study of various attributes, like principles, methods and processes, challenges, etc.
- **Four principles** form the foundation of the design thinking process.
- A design thinker needs to form his ideas and put them forward based on these principles.



Attributes (II)

Challenges: wicked problems

- **Wicked problems** are the challenges that are faced by the design thinkers
- They are supposed to be **ill-defined or tricky**
- In the case of ill-defined problems, the **problem statement** and **the solution** are **both unknown** at the beginning of the design thinking exercise

Attributes (III)

The Aha-Moment

- Moment when **the thinker suddenly finds his way clear of all obstructions**
 - This is the moment when the solution or a bright idea strikes the thinker's mind
- The focus on the solution grows clear after this moment and the final product or the **final solution is constructed hereafter**

Attributes (IV)

Design Methods

- **Every design discipline** makes **use** of a set of specific techniques, rules, and ways of doing things. These are called **design methods**
 - They include interviewing, creating user profiles, creating mind maps, creating prototypes for solving a problem and asking for answers to questions like five whys.
 - The ‘five whys’ is an iterative interrogative technique, which is used to explore the cause-and-effect relationships underlying a particular problem

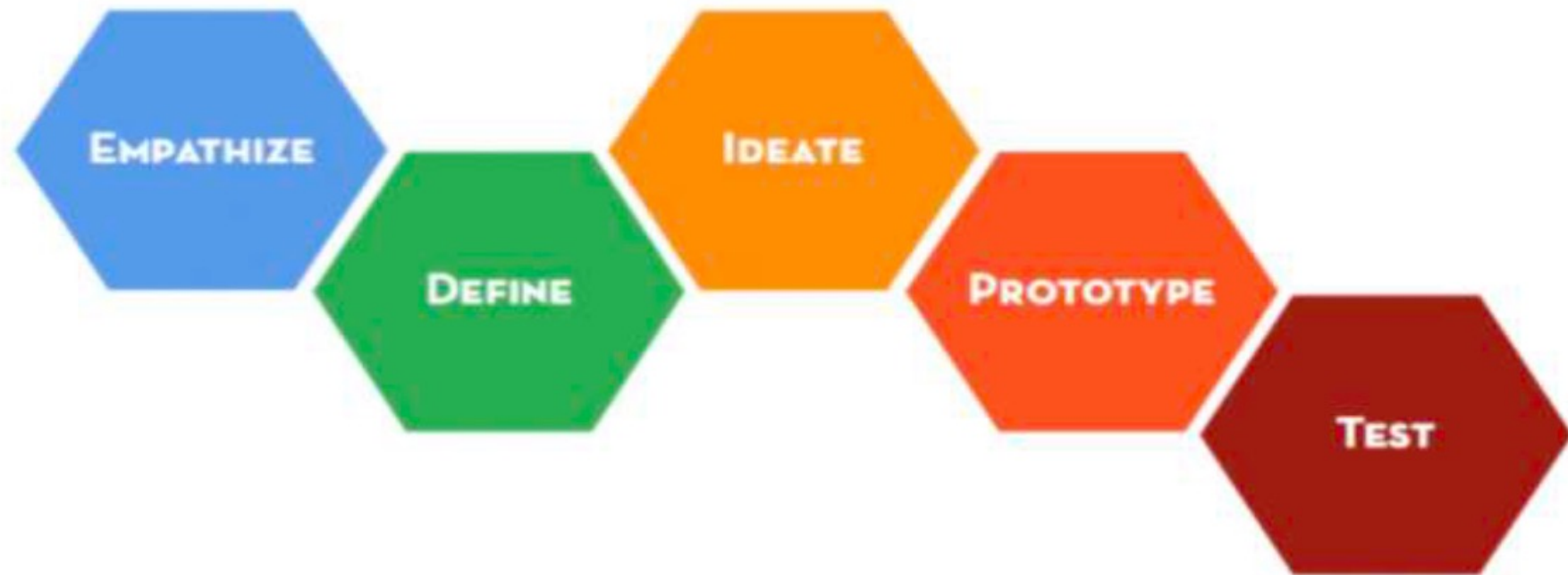
Attributes (V)

Use of analogies

- It is imperative for a design thinker to find logic even in ill-defined problems that contain obscure relationships
- This issue can be addressed using analogies

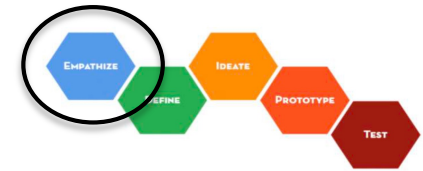
Attributes (VI)

The five-step process



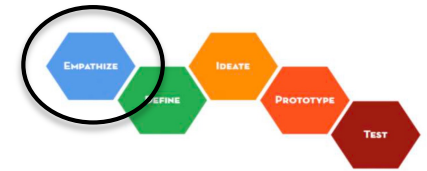
- Design thinkers are expected **not to think of the following steps when working on one step**
 - For example it is not recommended to think of solutions, when the problem is being defined

Empathize Stage (I)



- It involves **putting oneself into the shoes** of the customer or the **end-user** of our solution
 - We need to **understand the problems** faced by the **customer**
 - As design thinkers, we **need to empathize** with the **customer**
 - It involves the **process of analysis**
- **Empathy is the centerpiece of a human-centered design process**
- Observe what people do and how
 - It **gives feedback** about **what** they **think** and **feel**
 - Helps infer **intangible meaning** of those experiences
 - **uncover insights** → **give direction of innovative solutions**

Empathize Stage (II)

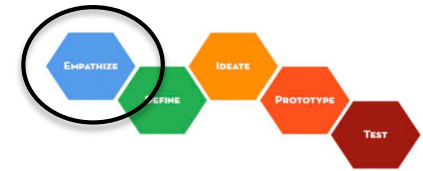


The interviewer:

- must **brainstorm beforehand** and be fully prepared for the interview
- provide **open questions**
- must have **plenty of ‘why’** questions.

The **themes** of the questions must **not be intermingled**.

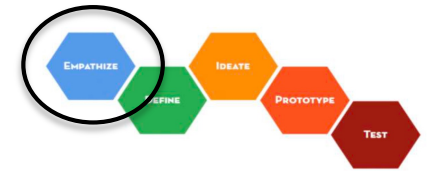
Empathize Stage (III)



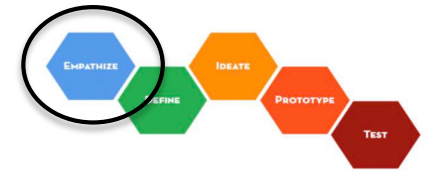
How to empathize

- **Observe** (in addition to interviews)
- **Engage** (conversation-style, free, “why?”)
- **Watch** and **listen** (“user in action”)
- Get all the information out of your head and onto a **wall** where you can **start to make connections**
- This is the **beginning** of the **synthesis process** which leads to the next stage

Empathize Stage (IV)

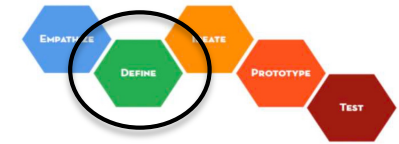


Empathize Stage (IV)



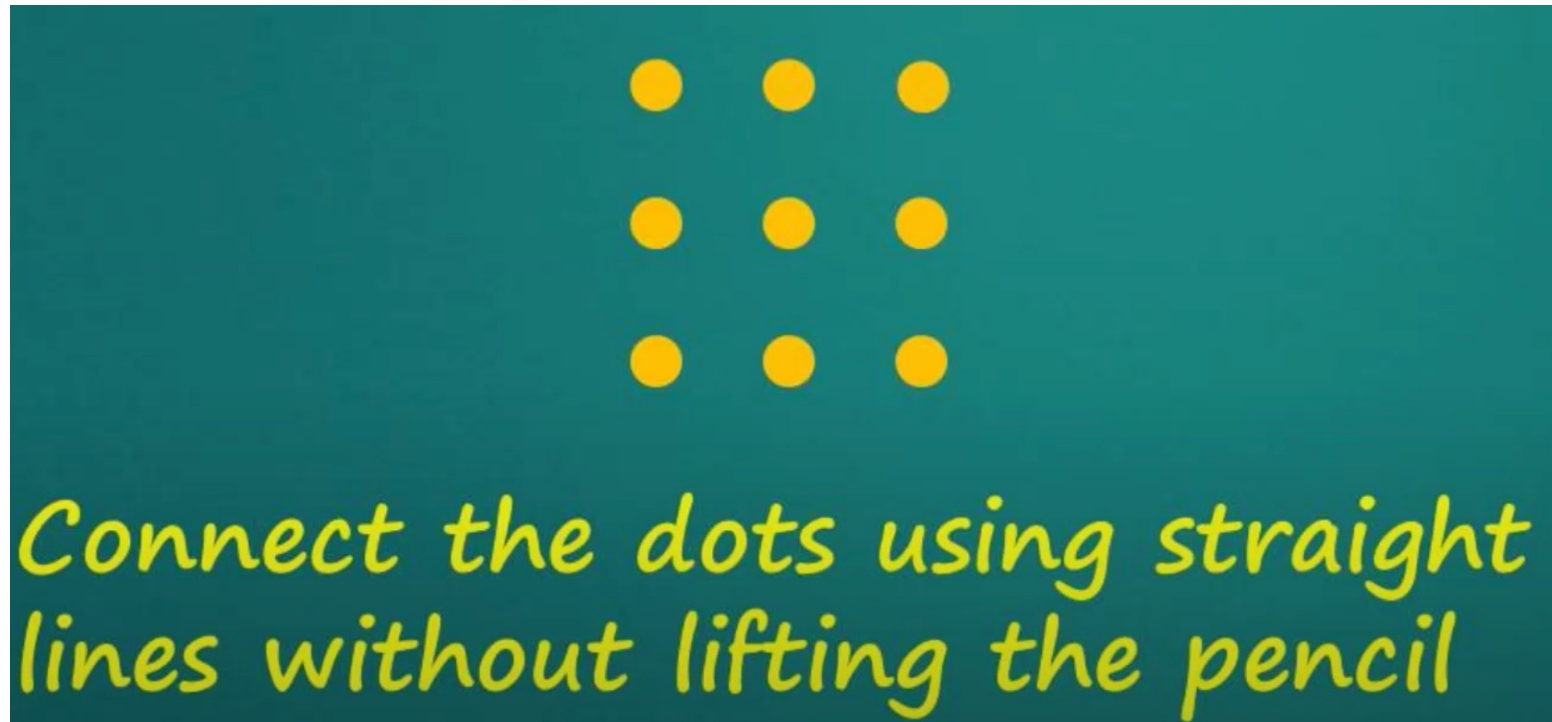
“To create meaningful innovations, you need know your users and care about their lives”

Define Stage (I)

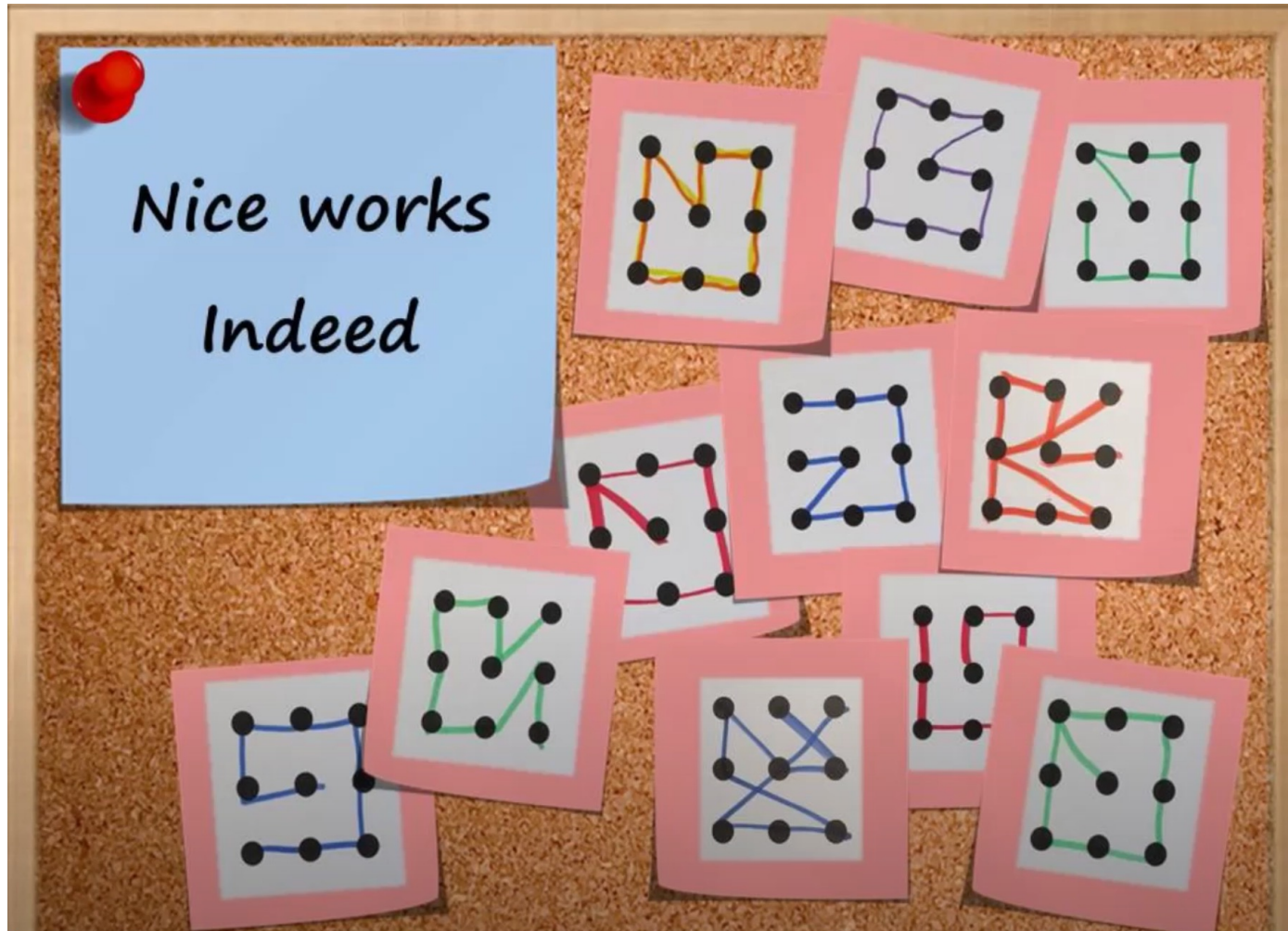


- It is time to define our problem → **problem statement**
- Information gathered → **coherent single statement**
- We have learnt the **problems** and **context** of our customers
 - User, needs, insights → **actionable statement**
- **Synthesis process** → bringing **clarity** and **focus** → **narrowing focus yields greater quantity/quality solutions**

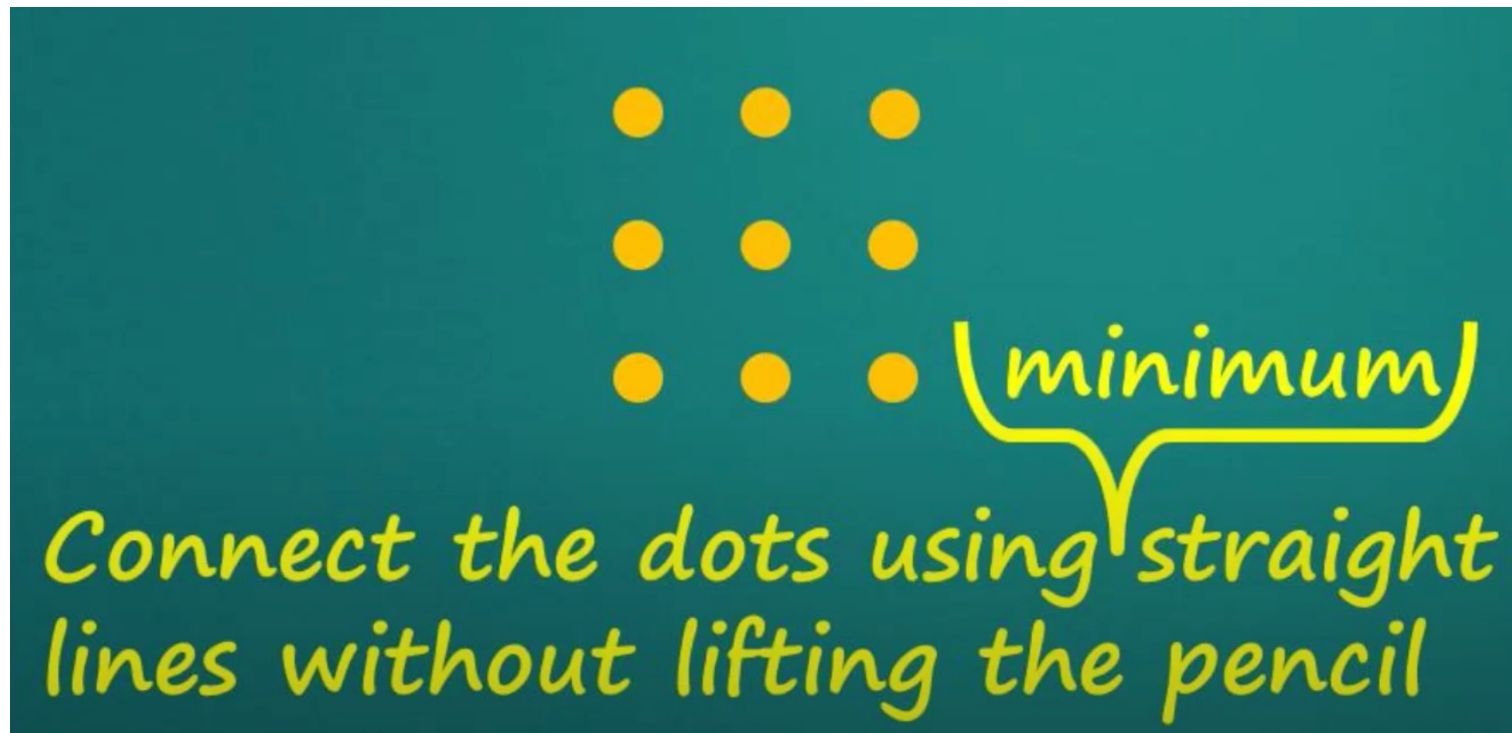
Creativity exercise: 9 points problem



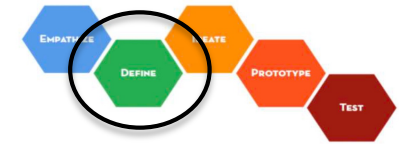
Creativity exercise: 9 points problem



Creativity exercise: 9 points problem



Define Stage (II)

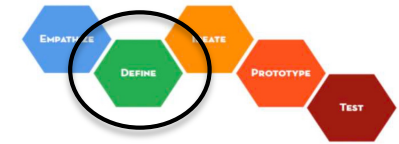


Problem statement → Think of ‘**How might we?**’ questions

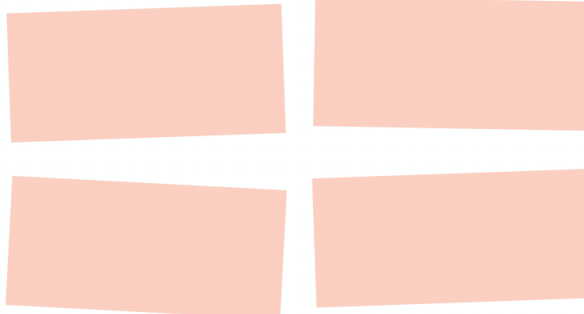
– How might we generate the ‘how might we’ questions?

- Amplify the good
- Eliminate the bad
- Explore the opposite (*turn problems into opportunity*)
- Question the assumptions
- Identify the unexpected (*not mentioned*) resources
- Create an analogy
- Break the problem into pieces (*analysis/synthesis*)

Define Stage (III)



HOW MIGHT WE..?

<i>What is the problem you want to solve?</i>	<i>Key Insights about the user</i> 
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HOW MIGHT WE HELP

Our user

SOLVE

The problem

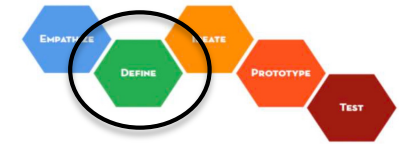
KNOWING THAT HE/SHE

?

needs/desires/insights we found out about the user

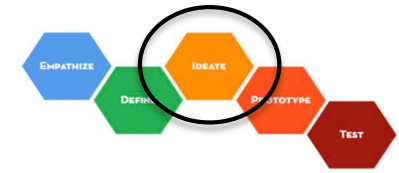
Picture from Design Thinking Playbook: <http://www.catalinacatana.com/wp-content/uploads/2018/10/Design-Thinking-playbook.pdf>

Define Stage (IV)



“Framing the right problem is the only way to create the right solution”

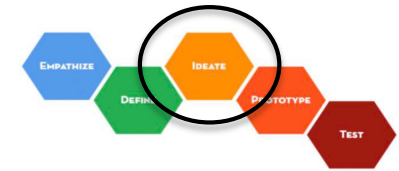
Ideate Stage (I)



- At this stage a design thinker is supposed to bring to the table **as many ideas as possible**
- While brainstorming for ideas, it is not checked if the idea is **possible, feasible, viable** or not.
- Combine **rational thoughts** with **imagination**
- All solutions suggested by design thinkers are brought to the table and **thought over**
- Ideation techniques: **prototyping, mindmapping, bodystorming, sketching, ...**

Ideate Stage (II)

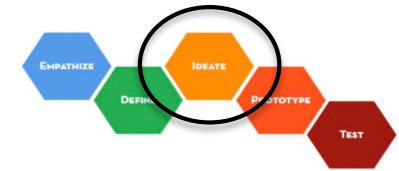
Rule for brainstorming



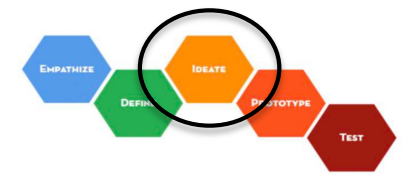
- Only **one conversation** is allowed at a time
 - No other person must intervene when an idea is being given.
- Focus must be on the **quantity** and not on **quality**
 - In this step, the group must have large number of ideas
- Think **out of the box**
 - Wild ideas must be encouraged even if they invoke plain humor or seem impossible.
- The group leader must **defer judgment**
 - The fellow thinkers also need to suspend judgment
 - Separate generation of ideas from evaluation of ideas

Ideate Stage (III)

“Tool 1”: Mind Map



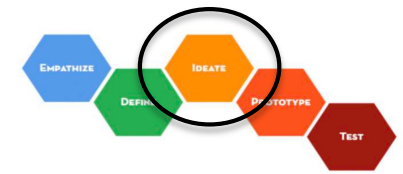
- Mind map is a diagram that helps to observe and study information in a **visual manner**
- It is created **around a single problem statement** and **all the ideas** to solve the problem **are written around it**
- The ideas can be represented as text, images, trees, and even smaller mind maps.
- It is also known by the name of **spider diagram**
- It is a **well-structured** organized diagram meant to **aid the thinking process** and to **streamline the analysis and synthesis processes**



Ideate Stage (IV)

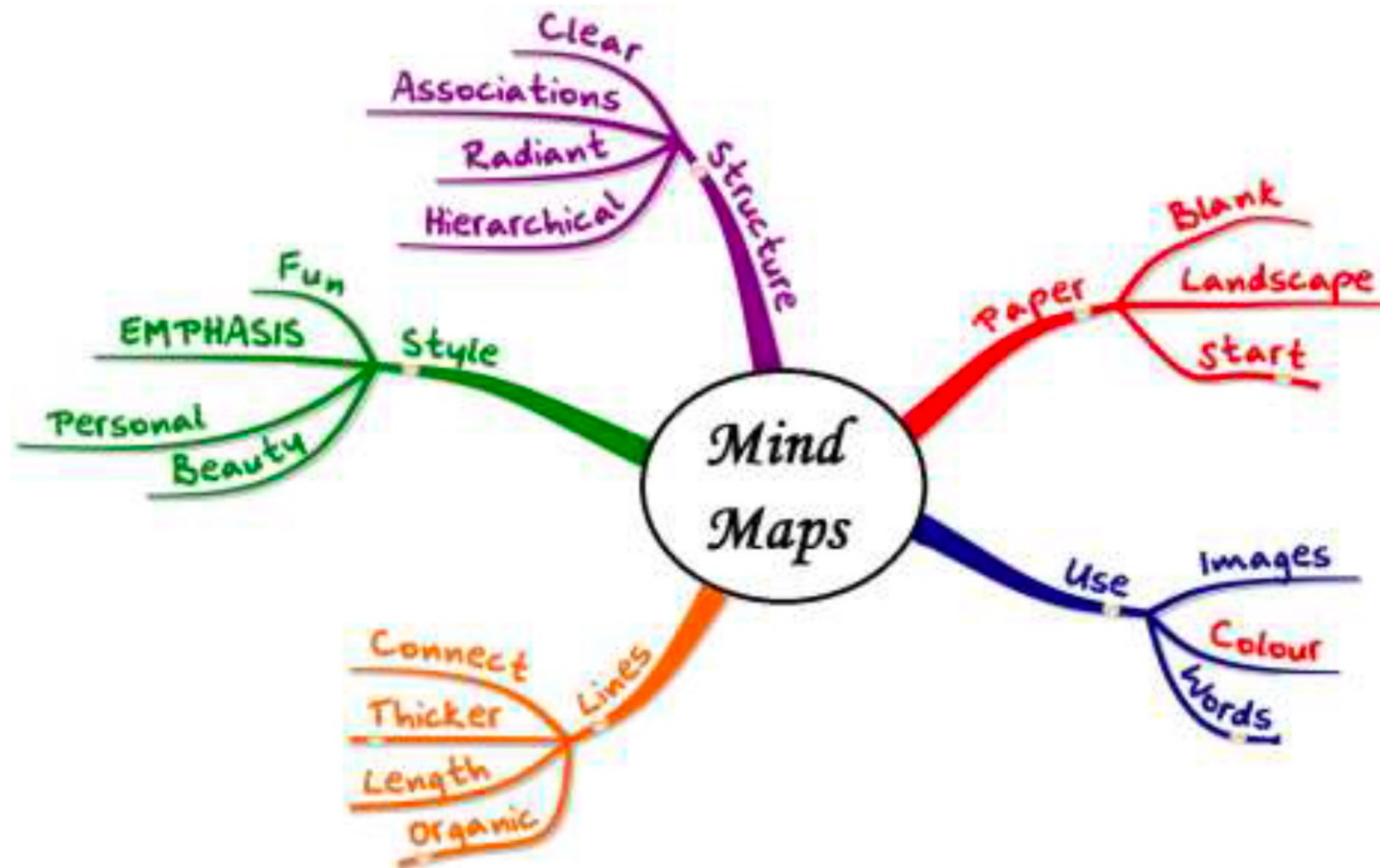
Guidelines for Mind Maps (Bullet list)

- Begin with the problem statement at the center of a blank white page
- Use images, different colors, symbols, caricatures, *abbr.* and codes to depict your ideas
- **Keywords** must replace long statements.
- Each and every word written in the mind map must be connected to the central hub by some or other line or set of lines
- Use multiple **colors** for **visual stimulation**
- Use radial hierarchy
- Make use of emphasis, *italics*, and underlines to stress on a point



Ideate Stage (V)

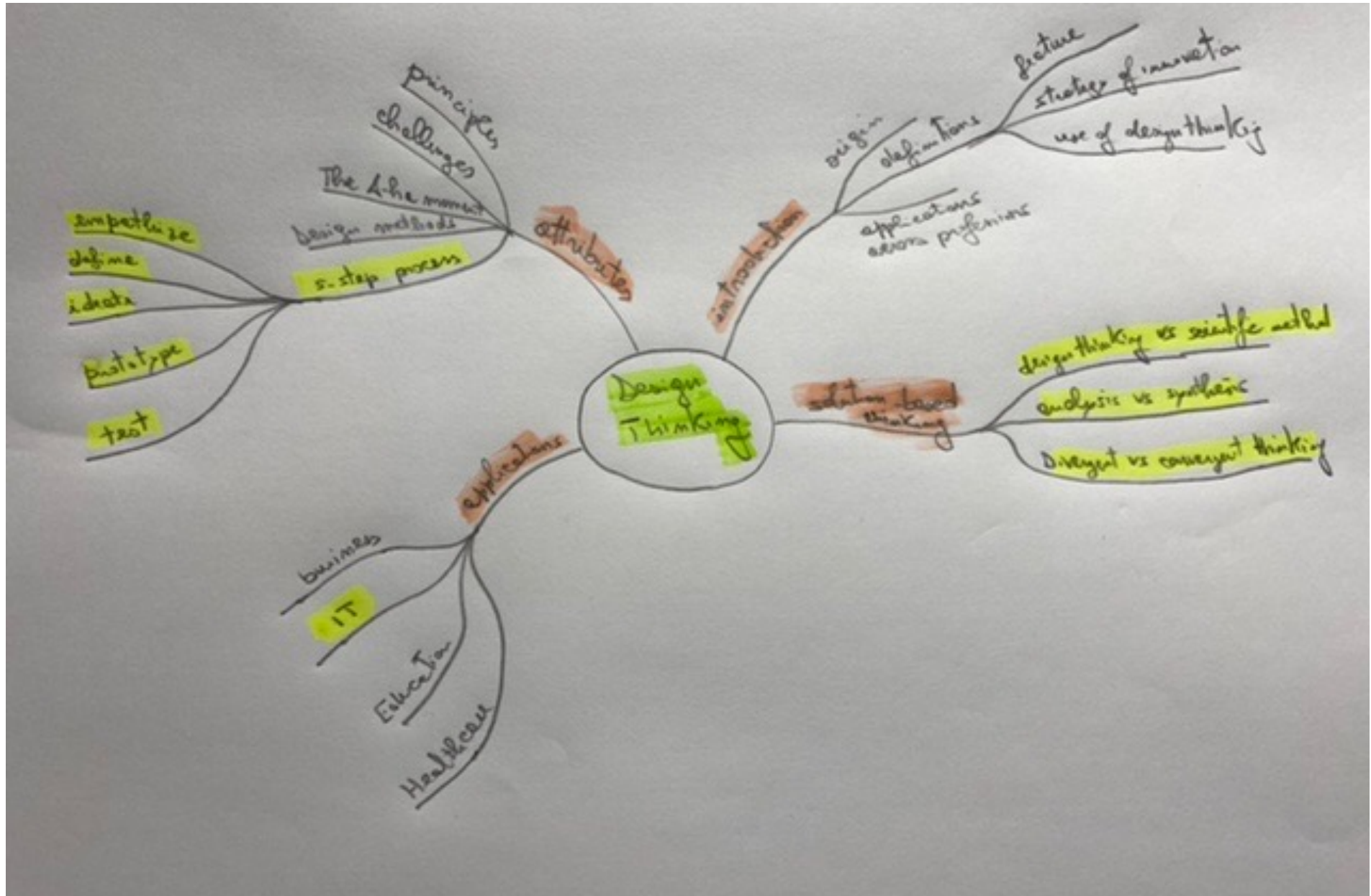
Guidelines for Mind Maps (Mind Map)



Exercise: “Today’s class outline”

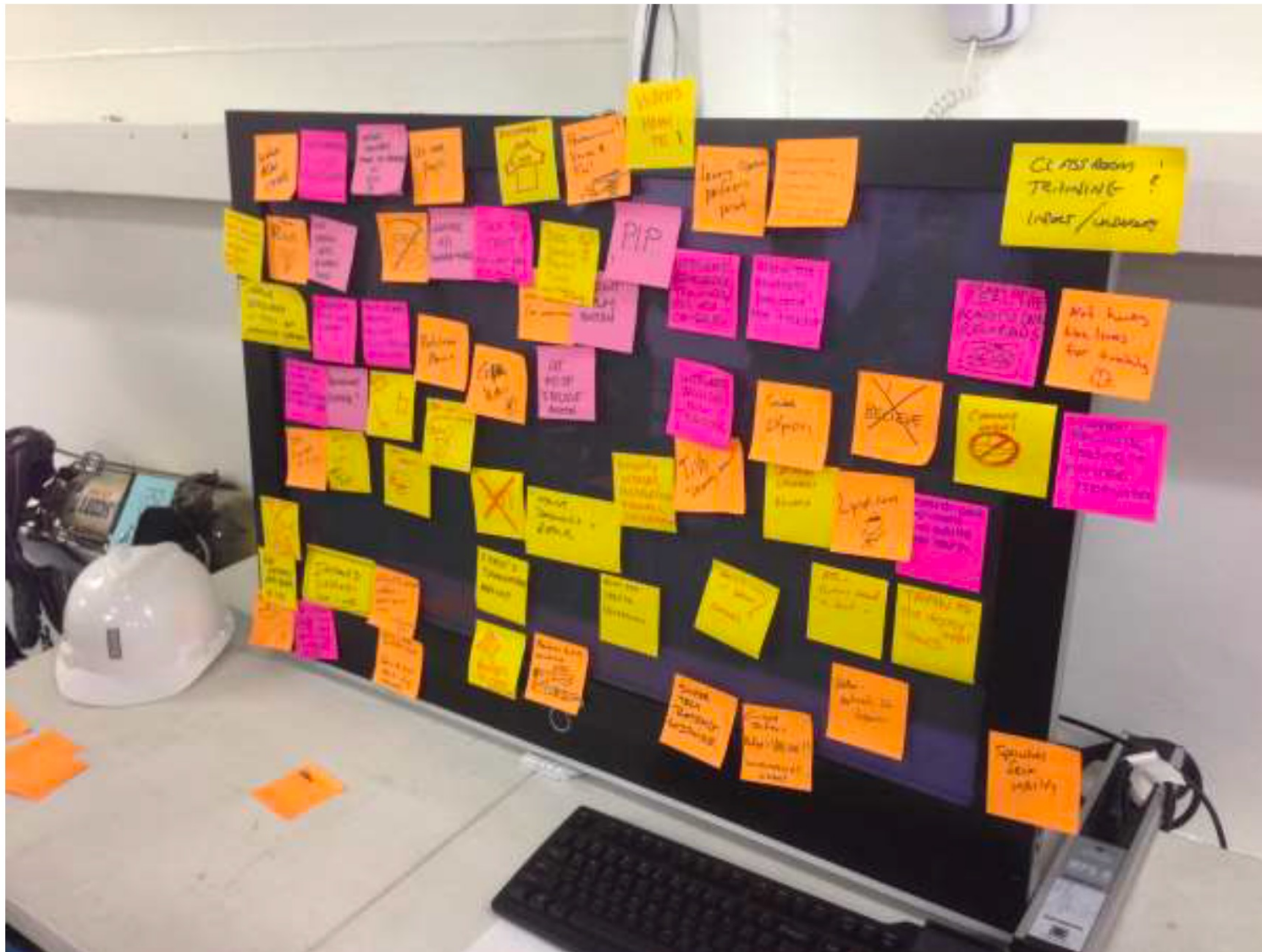
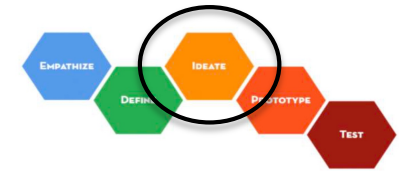
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- Design thinking and IT industry the agile era

Mind map: "Today's class"



Ideate Stage (VI)

“Tools 2”: Whiteboards

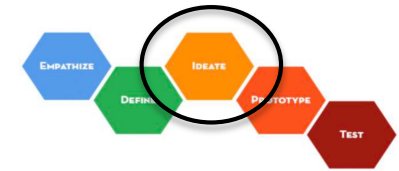


Ideate Stage (VII)

Transition to the next stage

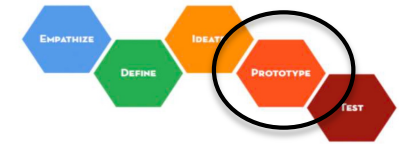
- **Selection process**
- As a team we designate three **voting criteria**
- Some examples (but they could be different):
 - 1) *“the most likely to delight”*
 - 2) *“the rational choice”*
 - 3) *“the most unexpected”*
- Vote and carry the **two** or **three ideas** that receive the most votes forward into **prototyping**
 - **Preserve innovation potential** by carrying forward multiple ideas instead of a single one

Ideate Stage (VIII)



“It’s not about coming up with the ‘right’ idea, it’s about generating the broadest range of possibilities”

Prototype Stage (I)



This step deals with:

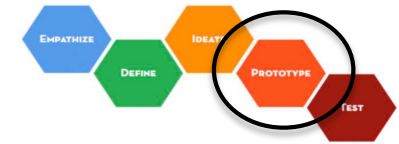
- building the ideas
- checking for their feasibility to arrive at the final solution.

In this step we take care of:

- Creation of experience (*for user, small scale, tangible*)
- Getting feedback (*from user*)
- Iteration (*through Empathize, Define, Ideate if necessary*)

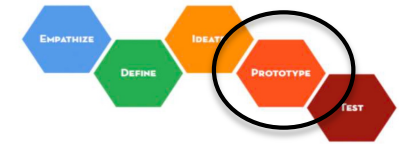
End users comes into play

Prototype Stage (II)



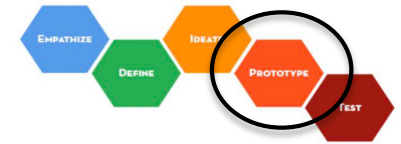
- Build the prototype → **iterative generation**
- **Don't waste too much of time/money** on building a single prototype
- The prototypes must be **built for and with the end user in mind**
- The prototype must create **an experience for the user**
- Think of **open questions** that the user can shoot towards you when he experiences the prototype
- There is no **value in the prototype** in case the **user** does not feel comfortable and satisfied with it

Prototype Stage (III)



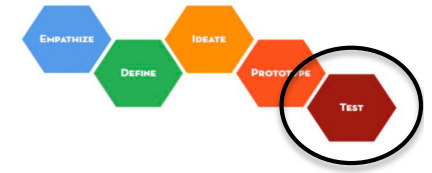
- Once the prototype has been developed
 - Take the end user through the prototype and let him/her experience it completely.
 - Make the user speak about his moment-by-moment experience.
 - Capture the minute details of the experience.
 - Try to actively observe and enthusiastically engage with the user during the experience.
- Once the experience is over, follow up with the user who had the experience with a set of questions

Prototype Stage (IV)



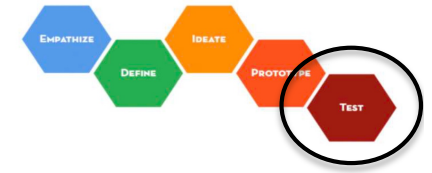
“Build to think and test to learn”

Test Stage (I)



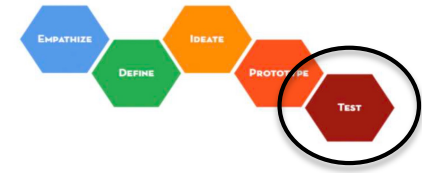
- The **idea** that seems the **best** according to the feedback of the customers and end users in the prototype phase will be executed
- **Full scale testing** to understand what actually works and what does not.
- This step can be:
 - the **most rewarding**, if the prototypes succeed to give positive results,
 - the **most annoying**, if the prototype fails.
- After testing
 - the entire process of design thinking **may have to be repeated**.
 - If the end user approves the solution, then the process of design thinking **stops here**.

Test Stage (II)



- Why test
 - refine prototype and solutions
 - learn more about users
 - refine problem statement
- How to test:
 - Show don't tell
 - Create experiences
 - Ask users to compare

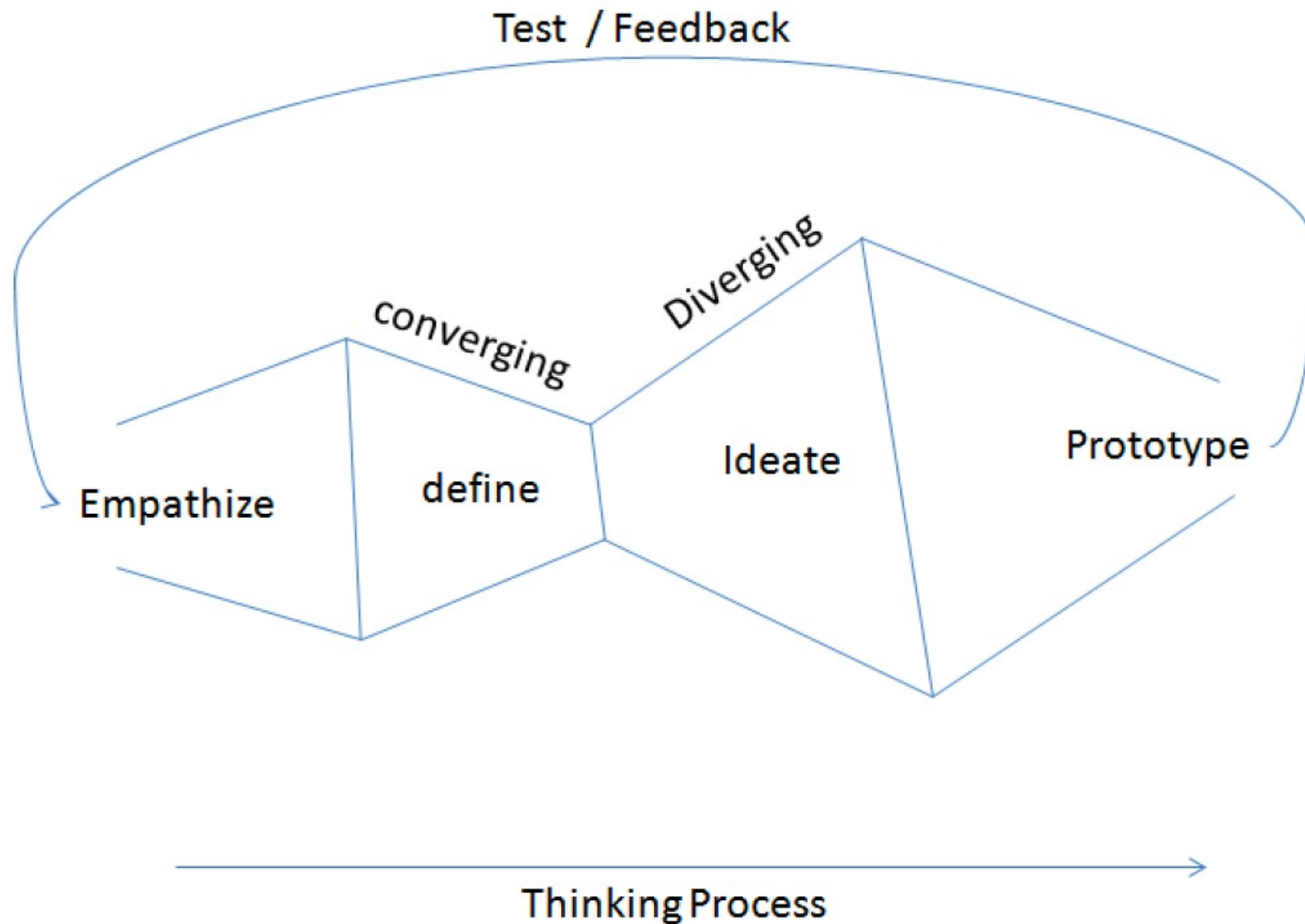
Test Stage (III)



“Testing is an opportunity to learn about your solution and your user”

Iterative process

Putting everything together



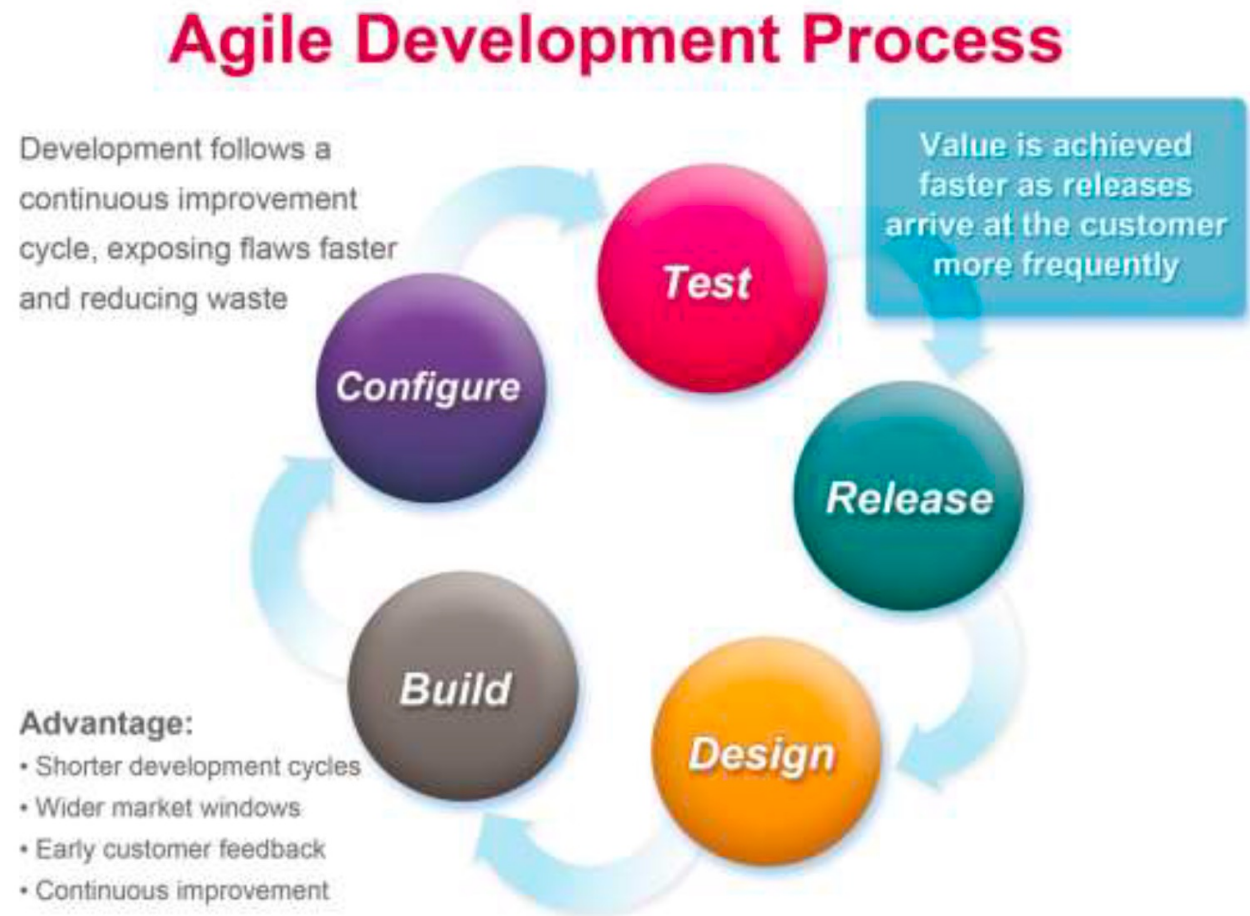
Design thinking process: a simple example

Look at the video below

[https://www.youtube.com/watch?v= r0VX-aU T8](https://www.youtube.com/watch?v=r0VX-aU_T8)

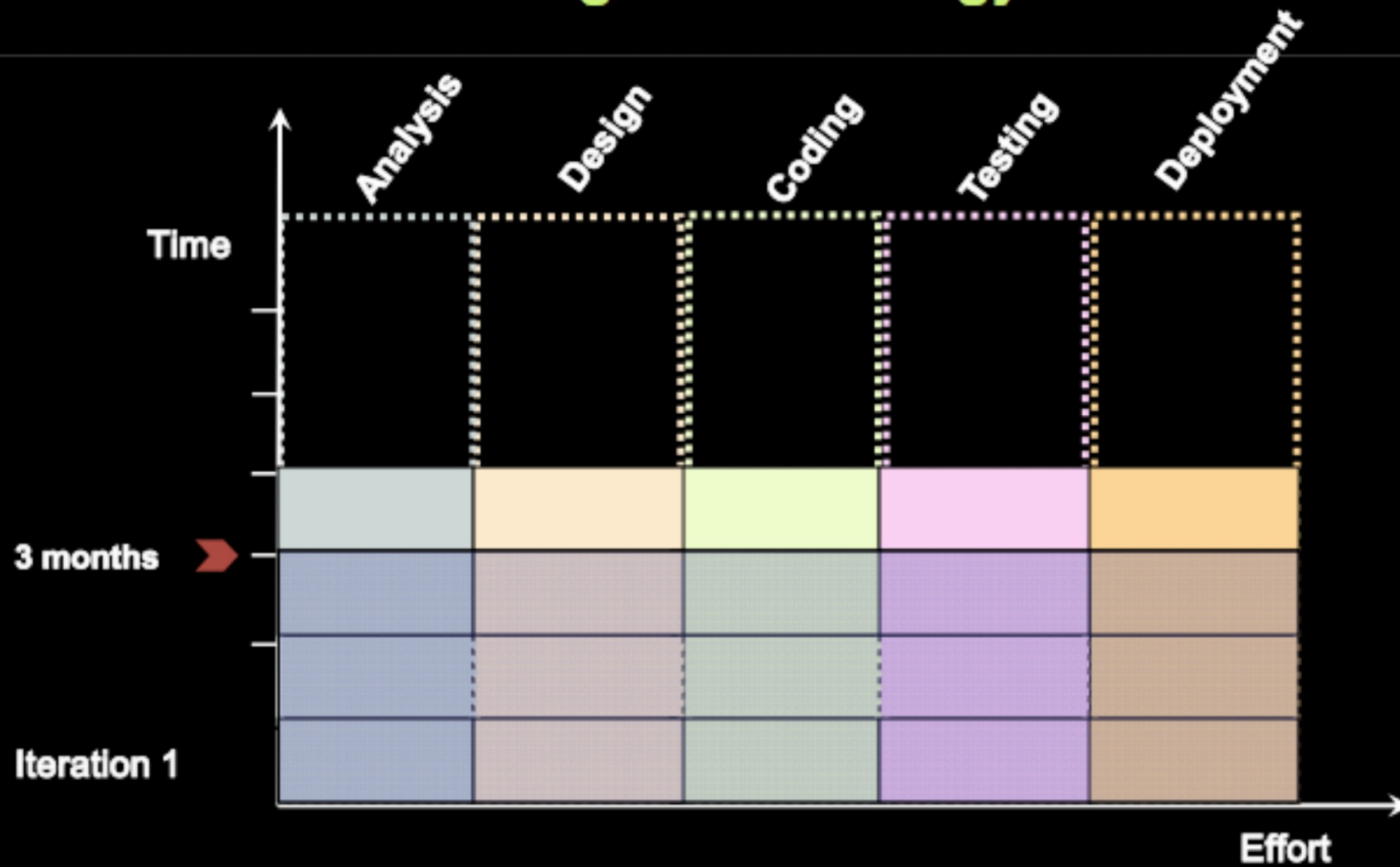
Design thinking & the IT industry in the agile era (I)

The IT industry is working mostly on Agile Methodologies



Principle #1: Our highest priority is to satisfy the **customer** through **early** and **continuous** delivery of **valuable** software

Agile Methodology



Design thinking & the IT industry in the agile era (II)

In such a scenario:

- *design thinking helps to solve the problems and address the exact needs of the customer*
- *Solving problems requires intuitive thinking and understanding of users needs*
- *No theories, no hypotheses*
- *After problem statement, designers brainstorm on ideas*
- *Solutions that are developed are sent to users for feedback*
- *Based on the feedback, the process can be reiterated*

Design thinking & the IT industry in the agile era (III)

The advantages of using design thinking in IT industry for software development are:

- The solutions are prototyped
- The results are verified
- The best solutions are accepted
- The solutions are experienced by the client before approval
- Short iterations are possible to improve the user experience
- Small cross-functional teams
- Incremental delivery is possible
- Fast feedback helps the designers and developers
- Continuous improvement is possible

Design thinking & the IT industry in the agile era (IV)

*Today the concept of **design thinking** is central to the **process of developing software solutions**, which target the **exact needs** of the **customer** and have the **flexibility** of getting modified in an **iteration process** based on the **feedback** given by the **client**.*

Conclusions

Core values of Design Thinking

- Human-centered design, empathy
- Experimentation, prototyping, make it
- Learn from failure
- A bias towards action
- Show, don't tell
- Power of iteration
- Embrace Ambiguity

Homework: IDEO Shopping Cart

- Take a few minutes to watch the following video about the IDEO Shopping Cart:

<https://www.youtube.com/watch?v=M66ZU2PClcM>

- Highlight the 2 aspects (**2 single words**) of the process you believe are the most relevant for a successful result!
 - Write down **all** the most relevant aspects
 - Select the top **2**

Feedback form

- There will be a ***feedback form*** open throughout the entire course
 - Please provide any comment about:
 - Pros
 - Cons
 - Aspects that were not clear enough during the class
 - Any other feedback you think can be relevant for the course
- Feedback will be collected anonymously
- The feedback form is available here:
<https://docs.google.com/forms/d/e/1FAIpQLScOmS0cpVoXbpABTkVHscLb1z--11ScOh1MSyUnk9u38MUWsA/viewform>